

BILL OF MATERIALS			
Part	Quantity	Price	Notes
Limit switch	5	40	
ESP32	2	560	(-280 for reuse from Basics of Mechatronics)
Diode	5	20	(-20 for reuse from Basics of Mechatronics)
Cooling Fan	1	50	
Color Sensor	1	325	
Vaccum Pump	1	350	Suction system to hold the parts
Stepper Motor	2	1000	High load carrying capacity + easy to identify position
Stepper Driver	2	200	
Timing belt system	2	1400	Cheap & Reliable
Motor + Encoder	1	500	For closed loop PID
Power Screw	1	1000	Easy to maintain
Laser Printing	1	2000	Cheap, fast , and easy manufacturing method
3D printing	1	1500	for non standard parts
Misc	1	2000	Buffer
Sum		10945	EGP